

Introduction

Budgets and the Budgeting Cycle

A *budget* is:

- a. The quantitative expression of a proposed plan of action by management for a specified period and
- b. An aid to coordinate what needs to be done to implement that plan.

A budget is a comprehensive formal management plans expressed in quantitative terms, describing the expected operations of an organization over some future time period. A budget deals with a specific entity, covers a specific future time period and is expressed in quantitative terms.

A budget generally includes both financial and nonfinancial aspects of the plan, and it serves as a blueprint for the company to follow in an upcoming period. A financial budget quantifies management's expectations regarding income, cash flows, and financial position. Just as financial statements are prepared for past periods, financial statements can be prepared for future periods—for example, a budgeted income statement, a budgeted statement of cash flows, and a budgeted balance sheet. Underlying these financial budgets are nonfinancial budgets for, say, units manufactured or sold, number of employees, and number of new products being introduced to the marketplace.

Human Factors in Budgeting

The success of a budget program also depends on the degree to which top management accepts the budget program as a vital part of the company's activities and the way in which top management uses budgeted data. If a budget program is to be successful, it must have the complete acceptance and support of the persons who occupy key management positions. If lower or middle management personnel sense that top management is lukewarm about budgeting, or if they sense that top management simply tolerates budgeting as a necessary evil, then their own attitudes will reflect a similar lack of enthusiasm. Budgeting is hard work, and if top management is not enthusiastic about and committed to the budget program, then it is unlikely that anyone else in the organization will be either.

In administering the budget program, it is particularly important that top management not use the budget to pressure or blame employees. Using budgets in such negative ways will breed hostility, tension, and mistrust rather than cooperation and productivity. Unfortunately, the budget is too often used as a pressure device and excessive emphasis is placed on "meeting the budget" under all circumstances. Rather than being used as a

weapon, the budget should be used as a positive instrument to assist in establishing goals, measuring operating results, and isolating areas that need attention.

The human aspects of budgeting are extremely important. The remainder of the chapter deals with technical aspects of budgeting, but do not lose sight of the human aspects. The purpose of the budget is to motivate people and to coordinate their efforts. This purpose is undermined if managers become preoccupied with the technical aspects or if the budget is used in a rigid and inflexible manner to control people.

Activity Based Budgeting

Activity-based budgeting (ABB) is **a system that records, researches, and analyzes activities that lead to costs for a company**. Every activity in an organization that incurs a cost is scrutinized for potential ways to create efficiencies. Budgets are then developed based on these *results*. Like the name implies, an activity-based budget is created using the activities that drive costs. The main objective of activity-based budgeting is **to cut down on costs**. It's an in-depth analysis of expenses and how they may affect a company's performance

Activity-Based Budgeting – Advantages

Relative to other budgeting methods, ABB allows you to see exactly what the associated costs are for each operational activity. It also helps to further break down these costs to determine what can be hurting the profitability of a company. While other methods of budgeting look at the costs of inputs to perform activities, ABB looks at the outputs that drive costs. In doing so, management can better evaluate different business units relative to each other and allocate [capital](#) where they deem to be most profitable.

Activity-Based Budgeting – Disadvantages

The biggest disadvantage of implementing ABB is that it is more costly and time-consuming to implement than other budgeting methods. As all costs associated with a business activity are tracked, all technical details must be recorded as they occur.

Budgetary control theory

This theory explains the relationship between budgetary control and financial performance of organization. According to this theory, a budgetary control process is a tool used by an organization as a framework for revenue and expenditure allocation.

Relevance of budget control

Budget controls are necessary to ensure that a government does not spend more than the amount legally appropriated by its governing body. By establishing clear spending boundaries, budget controls also promote accountability and bolster trust throughout the organization.

What are the three advantages of budget control?

Steps in Budgetary Control System

1. Preparing the budget
2. Communicating the budget
3. Measuring result
4. Comparing result & analyzing deviations
5. Reporting the result and taking corrective actions
6. Revising the budget

Advantages of Budgetary Control

The budgetary control system help in fixing the goals for the organization as the whole and concerted efforts are made for its achievements.

Some of the advantages of budgetary control are:

Maximization of Profits

The budgetary control aims at the maximization of profits of the enterprise. To achieve this aim, proper planning and coordination of different functions are undertaken. There is proper control over various capital and revenue expenditures. The resources are put to the best possible use.

Co-ordination

The working of different departments and sectors is properly coordinated. The budgets of different departments have a bearing on one another. The coordination of various executives and subordinates is necessary for achieving budgeted targets.

Specific Aims

The plans, policies and goals are decided by the top management. All efforts are put together to reach the common goal, of the organization. Every department is given a target to be achieved. The efforts are directed towards achieving some specific aims. If there is no definite aim then the efforts will be wasted in pursuing different aims.

Tool for Measuring Performance

By providing targets to various departments, the budgetary control provides a tool for measuring managerial performance. The budgeted targets are compared to actual results and deviations are determined. The performance of each department is reported to the top management. This system enables the introduction of management by exception.

Economy

The planning of expenditure will be systematic and there will be an economy in spending. The finances will be put to optimum use. The benefits derived from the concern will ultimately extend to the industry and then to the national economy. The national resources will be used economically and wastage will be eliminated.

Determining Weaknesses

The deviations in budgeted and actual performance will enable the determination of weak spots. Efforts are concentrated on those aspects where performance is less than the stipulated.

Corrective Action

The management will be able to take corrective measures whenever there is a discrepancy in performance. The deviations will be regularly reported so that necessary action is taken at the earliest. In the absence of a budgetary control system, the deviations can be determined only at the end of the financial period.

Consciousness

It creates budget consciousness among the employees. By fixing targets for the employees, they are made conscious of their responsibility. Everybody knows what he is expected to do and he continues with his work uninterrupted.

Reduces Costs

In the present day, competitive world budgetary control has a significant role to play. Every businessman tries to reduce the cost of production for increasing sales. He tries to have those combinations of products where profitability is more.

Introduction of Incentive Schemes

The budgetary control system also enables the introduction of incentive schemes of remuneration. The comparison of budgeted and actual performance will enable the use of such schemes.

Limitations of Budgetary Control

Despite many good points of budgetary control, there are some limitations to this system.

Uncertain Future

The budgets are prepared for the future period. Despite best estimates made for the future, the predictions may not always come true. The future is always uncertain and the situation which is presumed to prevail in future may change. The change in future conditions upsets the budgets which have to be prepared on the basis of certain assumptions. The future uncertainties reduce the utility of a budgetary control system.

Budgetary Revision Required

Budgets are prepared on the assumptions that certain conditions will prevail. Because of future uncertainties, assumed conditions may not prevail necessitating the revision of budgetary targets. The frequent revision of targets will reduce the value of budgets and revisions involve huge expenditures too.

Discourage Efficient Persons

Under the budgetary control system, the targets are given to every person in the organization. The common tendency of people is to achieve the targets only. There may be some efficient persons who can exceed the targets but they will also feel content by reaching the targets. So budgets may serve as constraints on managerial initiatives.

Co-ordination Problem

The success of budgetary control depends upon the coordination among different departments. The performance of one department affects the results of other departments. To overcome the problem of co-ordination a Budgetary Officer is needed. Every concern cannot afford to appoint a Budgetary Officer. The lack of coordination among different departments results in poor performance.

Conflict Among Different Departments

Budgetary control may lead to conflicts among functional departments. Every departmental head worries for his department goals without thinking of business goal. Every department tries to get the maximum allocation of funds and this raises a conflict among different departments.

Depends Upon Support of Top Management

Budgetary control system depends upon the support of top management. The management should be enthusiastic about the success of this system and should give full support for it. If at any time there is a lack of support from top management then this system will collapse.

Uncertainty and budgeting,

An uncertainty budget is an itemized table of components that contribute to the uncertainty in measurement results. It reveals important information that identifies, quantifies, and characterizes each source of uncertainty

In the budgeting process, there may be uncertainty at various levels.

Causes of uncertainty in the budgeting process

- **Customers:** They may decide to buy less than forecast, or they may buy more.
- **Products/services:** In the modern business environment, organizations need to respond to customers' rapidly changing requirements.
- **Inflation and movements in interest and exchange rates.**
- **Volatility in the cost of materials.**
- **Competitors:** They may steal some of an organization's expected customers, or some competitors' customers may change their buying allegiance.
- **Employees:** They may not work as hard as was hoped, or they may work harder.
- **Machines:** They may break down unexpectedly.
- **There may be political unrest (terrorist activity), social unrest (public transport strikes) or minor or major natural disasters (storms, floods).**

Types of budgetary systems

There are several budgeting systems an organization can use depending on the prevailing circumstances. These are:

Rolling budgets and flexible budgets

These are a way of trying to reduce the element of uncertainty in the plan. There are other planning methods which try to analyze the uncertainty such as probabilistic budgeting, sensitivity analysis (what if?), scenario planning and simulation.

These methods are suitable when the degree of uncertainty is quantifiable from the start of the budget period and actual results are not expected to go outside the range of these expectations.

Zero Based Budgeting

Zero-based budgeting (ZBB) is a budgeting technique in which all expenses must be justified for a new period or year starting from zero, versus starting with the previous budget and adjusting it as needed.

The process of zero-based budgeting starts from a "zero base," and every function within an organization is analyzed for its needs and costs. The budgets are then built around what is needed for the upcoming period, regardless of whether each budget is higher or lower than the previous one.

Zero-Based Budgeting vs Traditional Budgeting

Traditional budgeting calls for incremental increases over previous budgets, such as a 2% increase in spending, as opposed to a justification of both old and new expenses, as called for with zero-based

budgeting. Traditional budgeting also only analyzes only new expenditures, while ZBB starts from zero and calls for a justification of old, recurring expenses in addition to new expenditures. Zero-based budgeting aims to put the onus on managers to justify expenses and aims to drive value for an organization by optimizing costs and not just revenue.

Example of Zero-Based Budgeting

Suppose a construction equipment company implements a zero-based budgeting process calling for closer scrutiny of manufacturing department expenses. The company notices that the cost of certain parts used in its final products and outsourced to another manufacturer increases by 5% every year. The company can make those parts in-house using its workers. After weighing the positives and negatives of in-house manufacturing, the company finds it can make the parts more cheaply than the outside supplier.

Instead of blindly increasing the budget by a certain percentage and masking the cost increase, the company can identify a situation in which it can decide to make the part itself or buy the part from the external supplier for its end products.

What Are the Advantages of Zero-Based Budgeting?

As an accounting practice, zero-based budgeting offers a number of advantages including focused operations, lower costs, budget flexibility, and strategic execution. When managers think about how each dollar is spent, the highest revenue-generating operations come into greater focus. Meanwhile, lowered costs may result as zero-based budgeting may prevent the misallocation of resources that may happen over time when a budget grows incrementally.

What Are the Disadvantages of Zero-Based Budgeting?

Zero-based budgeting has a number of disadvantages. First, it is timely and resource-intensive. Because a new budget is developed each period, the time cost involved may not be worthwhile. Instead, using a modified budget template may prove more beneficial. Second, it may reward short-term perspectives in the company by allocating more resources to operations with the highest revenues. In turn, areas such as research and development, or those that have a long-term horizon, may get overlooked.

Incremental Budgeting

Incremental budgeting is the traditional budgeting method whereby the budget is prepared by taking the current period's budget or actual performance as a base, with incremental amounts then being added for the new budget period.

Continuous budget

Continuous budgeting is the process of continually adding one more month to the end of a multi-period budget as each month goes by. The continuous budgeting concept is usually applied to a twelve-month budget, so there is always a full-year budget in place

Periodic budget

A periodic budget, as the name suggests, involves dividing the annual budget into smaller periods. It is more useful to refer to pay periods since money coming in and out centers around them. To establish a periodic budget, you have to know the available amount of your cash assets when creating the budget.

Short Term and long Term Planning

Short-term planning is the process of deciding what objectives to pursue during a short, near-future period, usually one year, and what to do to achieve those objectives. The typical short-term budget covers one year and is broken down into monthly or quarterly units. Another method frequently used to prepare a short-term budget is the **continuous budget**. This kind of budget starts with an annual budget broken down into 12 monthly units. As each month arrives, it is dropped from the plan and replaced by a new month so that at any given time, the next 12 months are always shown. Thus, in a budgetary period covering January through December 20X4, when January 20x4 arrives, it would be dropped from the plan and replaced by January 20x5, thus creating a new budgetary period covering February 20x4 through January 20x5. Using this technique, a firm always has guidance for the full following year. When a continuous budget is not used, a firm will have guidance for only a month or two as it approaches the end of its budgetary period.

Long-term planning, also known as **strategic planning**, is the process of setting long-term goals and determining the means to attain them. Short-term planning is concerned with operating details for the next accounting period, but long-term planning addresses broad issues, such as new product development, plant and equipment replacement, and other matters that require years of advance planning. For example, short-term planning in the automotive industry would be concerned with which and how many of the current year's models to manufacture, while long-range planning would focus on new model development and major changes, as well as equipment replacements and modifications. The time frame for long-range planning may extend as far as 20 years in the future, but its usual range is from 2 to 10 years. An important part of long-term planning is the preparation of **the capital budget**, which details plans for the acquisition and replacement of major portions of property, plant, and equipment.

Principal Advantages of Budgeting

As noted earlier, *a budget* is a detailed plan expressed in quantitative terms that specifies how resources will be acquired and used during a specific period of time. The act of preparing a budget is known as *budgeting*. The use of budgets to control a firm's activities is called *budgetary control*.

Companies realize *many benefits* from a budgeting program. Among these benefits are the following:

- Requires periodic planning.
- Fosters coordination, cooperation, and communication.
- Provides a framework for performance evaluation.
- Means of allocating resources.
- Satisfies legal and contractual requirements.
- Created an awareness of business costs.

Periodic Planning (Formalization of Planning): The most obvious purpose of a budget is to quantify a plan of action. The development of a quarterly budget for a Sheraton Hotel, for example, forces the hotel manager, the reservation manager, and the food and beverage manager to plan for the staffing and supplies needed to meet anticipated demand for the hotel's services. To sum up, budgets forces managers to think ahead to anticipate and prepare for the changing conditions. The budgeting process makes planning an explicit management responsibility.

Coordination, Cooperation and Communication: Planning by individual managers does not ensure an optimum plan for the entire organization. Therefore, any organization to be effective, each manager throughout the organization must be aware of the plans made by other managers. In order to plan reservations and ticket sales effectively, the reservation manager for Ethiopian Air Lines must know the flight schedules developed by the airline's route manager. The budget process pulls together the plans of each manager in an organization. In a nutshell, a good budget process communicates both from the top down and from the bottom up. Top management makes clear the goals and objectives of the organization in its budgetary directives to middle and lower level managers, and also to all employees. Employees and lower level managers inform top-level managers how they can plan to achieve the objectives.

Performance Evaluation or Framework for Judging Performance: Budgets are estimates of future events, and as such they serve as estimates of acceptable performance. Comparing actual result against budgeted results helps managers to evaluate the performance of individuals, departments, or entire companies.

Budgets are generally a better basis for judging actual results than is past performance. The major drawback of using historical results for judging current performance is that inefficiencies may be concealed in the past

performance.

Means of Allocating Resources: Because we live in a world of limited resources, virtually all individuals and organizations must ration their resources. The rationing process is easier for some than for other. Each person and each organization must compare the costs and benefits of each potential project or activity and choose those that result in the most appropriate resource allocation decision.

Generally, organizations resources are limited, and budgets provide one means of allocating resources among competing uses. The city of Addis Ababa, for example, must allocate its revenue among basic life services (such as police and fire protection), maintenance of property and equipment (such as city streets, parks and vehicles) and other community services (such as programs to prevent alcohol and drug abuse).

Legal and Contractual Requirements: Some organizations are required to budget because of legal requirements. Others commit themselves to budgeting requirement when signing loan agreements or other operating agreements. For example, a bank may require a firm to submit an annual operating budget and monthly cash budget throughout the life of a bank loan. Local police department, for example, would be out of funds if the department decided not to submit a budget this year.

Cost Awareness. Accountants and financial managers are concerned daily about the cost implications of decisions and activities, but many other managers are not. Production managers focus on input, marketing manager's focuses on sales, and so forth. It is easy for people to overlook costs and cost-benefit relationships. At budgeting time, however, all managers with budget responsibility must convert their plans for projects and activities to costs and benefits. This cost awareness provides a common ground for communication among the various functional areas of the organization.

Components of Master Budget

The **master budget** is the total budget package for an organization; it is the end product that consists of all the individual budgets for each part of the organization aggregated into one overall budget for the entire organization.

The two major components of master budget are the **operating budget** and the **financial budget**.

Operating budget: It focuses on income statement and its supporting schedules. It is also called **profit plan**. However, such budget may show a budgeted loss, or can be used to budget expenses in an organization or agency with no sales revenues.

Financial budget: It focuses on the effects that the operating budget and other plans will have on cash. The usual master budget for a non-manufacturing company has the following components.

- 1. Operating budget includes:**
- 2. Financial budget include:**

- a. Sales budget
- b. Purchases budget
- c. Cost of goods sold budget
- d. Operating expense budget

- a. Capital budget
- b. Cash budget
- c. Budgeted balance sheets
- d. Budgeted statement of cash flows

In addition to the master budget there are countless forms of special budgets and related reports. For example, a report might detail goals and objectives for improvements in quality or customer satisfaction during the budget period.

Figure 3-1 Preparation of Master Budget (Non manufacturing Company)

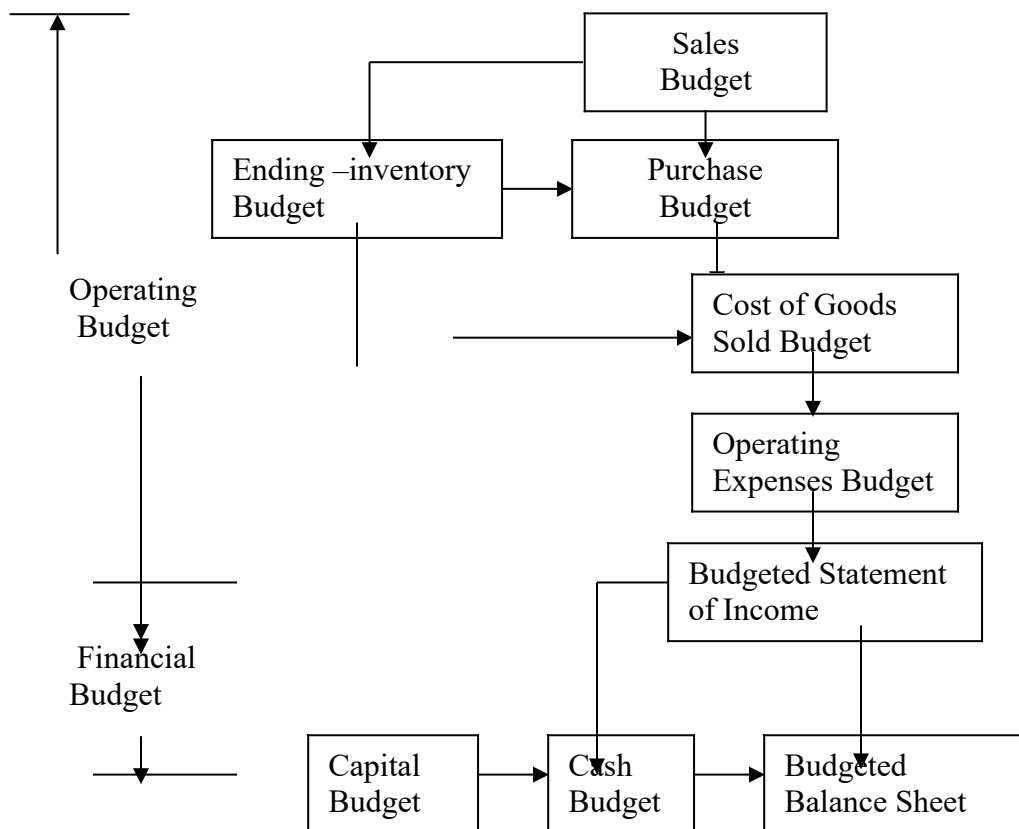


Exhibit 3-1 above show graphically the follow of process in development of the master budget for a non-manufacturing firm. The master budget example that follows should clarify the steps required to prepare the budget package. After studying the entire example, return to Exhibit 1-1 and follow the example through the flow diagram.

Operating Budget

The operating budget is composed of the income statement elements. A manufacturing business budgets both manufacturing and non-manufacturing activities. Below the various elements of the operating budget of a manufacturing firm have been discussed.

Sales Budget: The sales budget is the first budget to be prepared. It is usually the most important budget because so many other budgets are directly related to sales and are therefore largely derived from the sales budget. Inventory budgets, purchases budgets, personnel budgets, marketing budgets, administrative budgets, and other budget areas are all affected significantly by the amount of revenue that is expected from sales.

Sales budgets are influenced by a wide variety of factors, including general economic conditions, pricing decisions, competitor actions, industry conditions, and marketing programs. In an effort to develop an accurate sales budget, firms employ many experts to assist in sales forecasting. The sales budget is usually based on a sales forecast. A sales forecast is a prediction of sales under a given conditions. The objective in forecasting sales is to estimate the volume of sales for the period based on all the factors, both internal and external to the business that could potentially affect the level of sales. The projected level of sales is then combined with estimated of selling prices to form the sales budget. Sales forecasts are usually prepared under the direction of the top sales executive. Important factors considered by sales forecasters include:

a) Past patterns of sales: Past experience combined with detailed past sales by product line, geographical region, and type of customer can help predict future sales.

b) Estimates made by the sales force: A company's sales force is often the best source of information about the desires and plans of customers.

c) General economic conditions: Predictions for many economic indicators, such as gross domestic product and industrial production indexes (local and foreign), are published regularly. Knowledge of how sales relate to these indicators can aid sales forecasting.

d) Competitive actions: Sales depend on the strength and actions of competitors. To forecast sales, a company should consider the likely strategies and reactions of competitors, such as changes in their prices, products, or services.

f) Changes in the firm's prices: Sales can be increased by decreasing prices and vice versa. Planned changes in prices should consider effects on customer demand.

f) Changes in product mix: Changing the mix of products often can affect not only sales levels but also overall contribution margin. Identifying the most profitable products and devising methods to increases sales is a key part of successful management.

g) Market research studies: Some companies hire market experts to gather information about market conditions and customer preferences. Such information is useful to managers making sales forecasts and product mix decisions.

h) Advertising and sales promotion plans: Advertising and other promotional costs affect sales levels. A sales forecast should be based on anticipated effects of promotional activities.

Purchases Budget: After sales are budgeted, prepare the purchases budget. The total merchandise needed will be the sum of the desired ending inventory plus the amount needed to fulfill budgeted sales demand. The total need will be partially met by the beginning inventory; the remainder must come from planned purchases.

These purchases are computed as follows:

$$\text{Budgeted Purchases} = \text{Desired Ending inventory} + \text{Cost of Goods Sold} - \text{Beginning Inventory}$$

Budgeted cost of goods sold: For a manufacturing firm cost of goods sold is the production cost of products that are sold. Consequently, the cost of goods sold budget follows directly from the production budget. However, a merchandising firm has no production budget. The cost of goods sold budget comes directly from merchandise inventory and the merchandise purchases budget.

Operating Expense Budget: The budgeting of operating expenses depends on various factors. Month – to – month fluctuation in sales volume and other cost-drivers activities directly influence many operating expenses. Examples of expenses driven by sales volume include sales commissions and many delivery expenses. Other expenses are not influenced by sales or other cost-driver activity (such as rent, insurance, depreciation, and salaries) within appropriate relevant ranges and are regarded as fixed.

Budgeted Income Statement: The budgeted income statement is the combination of all of the preceding budgets. This budget shows the expected revenues and expenses from operations during the budget period.

A firm may have budgeted **non-operating items** such as interest on investments or gain or loss on the sale of fixed assets. Usually they are relatively small, although in large firms the amounts can be sizable. If non-operating items are expected, they should be included in the firm's budgeted income statement. Income taxes are levied on actual, not budgeted, net income, but the budget plan should include expected taxes; therefore, the last figure in the budgeted income statement is budgeted after tax net income.

Financial Budget

The second major part of the master budget is the financial budget, which consists of the capital budget, cash budget, ending balance sheet and the statement of changes in financial position. Although there are some

differences in operating budgets of manufacturing, merchandising and service firms, very little difference exists among financial budgets of these entities.

Capital expenditure budget: Capital budgeting is the planning of investments in major resources like plant and equipment, and other types of long-term projects, such as employee education programs. The **capital expenditure budget or capital budget** describes the capital investment plans for an organization for the budget period. It contains some of the most critical budgeting decisions of the organizations.

Cash budget: The **cash budget** is a statement of planned cash receipts and disbursements. The cash budget is composed of four major sections:

- i. **The receipts section:** It consists of a listing of all of the cash inflows, except for financing, expected during the budget period. Generally the major source of receipts will be from sales.
- ii. **The disbursement section:** It consists of all cash payments that are planned for the budget period. These payments will include inventory purchases, wages and salary payments and so on. In addition, other cash disbursements such as equipment purchases, dividends, and other cash withdrawals by owners are listed.
- iii. **The cash excess or deficiency section:** The cash excess or deficiency section is computed as follows:

Cash balance, beginning	x x x
Add receipts	<u>x x x</u>
Total cash available before financing	x x x
Less disbursements	<u>x x x</u>
Excess (deficiency) of cash available over disbursements	<u>x x x</u>

If there is a cash deficiency during any budget period, the company will need to borrow funds. If there is cash excess during any budget period, funds borrowed in previous periods can be repaid or the idle funds can be placed in short-term or other investments.

- iv. **The financing section:** This section provides a detail account of the borrowing and repayments projected to take place during the budget period. It also includes a detail of interest payments that will be due on money borrowed.

Budgeted Balance Sheet: The budgeted balance sheet, sometimes called the **budgeted statement of financial position**, is derived from the budgeted balance sheet at the beginning of the budget period and the expected changes in the account balance reflected in the operating budget, capital budget, and cash budget.

Budgeted Statement of Changes in Financial Position: The final element of the master budget package is the statement of changes in financial position. It has emerged as a useful tool for managers in the financial planning process. This statement is usually prepared from data in the budgeted income statement and changes between the estimated balance sheet at the beginning of the budget period and the budgeted balance sheet at the end of the budget period.

Preparing the Master Budget

The master budget is a network consisting of many separate but interdependent budgets. This network is illustrated in Exhibit 1-1. The master budget can be a large document even for a small organization. The simple example that follows on the next page for Blue Nile Company's give some indication of the potential size and complexity of the master budget of a business. The example illustrates a *fixed* or *static budget* prepared for a single expected level of activity. Flexible budgeting that involves various activity levels will be discussed later in the next unit.

Preparation of Master Budget (Manufacturing Company)

Example (2) Great Company manufactures and sells a product whose peak sales occur in the third quarter. Management is now preparing detailed budgets for 20x4- the coming year and has assembled the following information to assist in the budget preparation:

- 1) The company's product selling price is Br. 20 per unit. The marketing department has estimated sales as follows for the next six quarters.

	20x4 Quarters				20x5 Quarters	
	1	2	3	4	1	2
Budgeted sales in units	10, 000	30,000	40, 000	20, 000	15, 000	15, 000

- 2) Sales are collected in the following pattern: 70% of sales are collected in the quarter in which the sales are made and the remaining 30% are collected in the following quarter. On January1, 20x4, the company's balance sheet showed Br.90, 000 in account receivable, all of which will be collected in the first quarter of the year. Bad debts are negligible and can be ignored.
- 3) The company maintains an ending inventory of finished units equal to 20% of the next quarter's sales. The requirement was met on December 31, 20x3, in that the company had 2, 000 units on hand to start the New Year.

- 4) Fifteen pounds of raw materials are needed to complete one unit of product. The company requires an ending inventory of raw materials on hand at the end of each quarter equal to 10% of the following quarter's production needs of raw materials. This requirement was met on December 31, 20x3 in that the company had 21,000 pounds of raw materials to start the New Year.
- 5) The raw material costs Br.0.20 per pound. Raw material purchases are paid for in the following pattern: 50% paid in the quarter the purchases are made, and the remainder is paid in the following quarter. On January 1, 20x4, the company's balance sheet showed Br.25,800 in accounts payable for raw material purchases, all of which be paid for in the first quarter of the year.
- 6) Each unit of Great's product requires 0.8 hour of labor time. Estimated direct labor cost per hour is Br.7.50.
- 7) Variable overhead is allocated to production using labor hours as the allocation base as follows:

Indirect materials	Br.0.40
Indirect labor	0.75
Fringe benefits	0.25
Payroll taxes	0.10
Utilities	0.15
Maintenance	0.35

Fixed overhead for each quarter was budgeted at Br. 60,600. Of the fixed overhead amount, Br. 15,000 each quarter is depreciation. Overhead expenses are paid as incurred.

- 8) The company's quarterly budgeted fixed selling and administrative expenses are as follows:

	20X4 Quarters			
	1	2	3	4

Advertising	Br.20, 000	Br.20, 000	Br.20, 000	Br.20, 000
Executive salaries	55, 000	55, 000	55, 000	55, 000
Insurance	-	1, 900	37,750	-
Property taxes	-	-	-	18, 150
Depreciation	10, 000	10, 000	10, 000	10, 000

The only variable selling and administrative expense, sales commission, is budgeted at Br.1.80 per unit of the budgeted sales. All selling and administrative expenses are paid during the quarter, in cash, with exception of depreciation. New equipment purchases will be made during each quarter of the budget year for Br. 50, 000, Br. 40, 000, & Br.20, 000 each for the last two quarter in cash, respectively. The company declares and pays dividends of Br.8, 000 cash each quarter. The company's balance sheet at December 31, 20x3 is presented below:

ASSETS

Current assets:

Cash	Br. 42, 500
Accounts Receivable	90, 000
Raw Materials Inventory (21, 000 pounds)	4, 200
Finished Goods Inventory (2, 000 units)	<u>26, 000</u>
Total current assets	Br.162, 7 00

Plant and Equipment:

Land	Br.80, 000
Building and Equipment	700, 000
Accumulated Depreciation	<u>(292, 000)</u>

Plant and Equipment, net

488, 000

Total assets

Br.650, 700

Liabilities and Stockholders' Equity

Current liabilities:

Accounts payable (raw materials)	Br.25, 800
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Stockholders' equity:

Common stock, no par	Br.175, 000
Retained earnings	<u>449, 900</u>

Total stockholders' equity

624, 900

Total liabilities and stockholders' equity

Br.650, 700

The company can borrow money from its bank at 10% annual interest. All borrowing must be done at the beginning of a quarter, and repayments must be made at the end of a quarter. All borrowings and all repayments are in multiples of Br. 1,000.

The company requires a minimum cash balance of Br.40, 000 at the end of each quarter. Interest is computed and paid on the principal being repaid only at the time of repayment of principal. The company wishes to use any excess cash to pay loans off as rapidly as possible.

Instructions: Prepare a master budget for the four-quarter period ending December 31. Include the following detailed budget and schedules:

1. a) A sales budget, by quarter and in total
 - b) A schedule of budgeted cash collections, by quarter and in total
 - c) A production budget
 - d) A direct materials purchase budget
 - e) A schedule of budgeted cash payments for purchases by quarter and in total
 - f) A direct labor budget
 - g) A manufacturing overhead budget
 - h) Ending finished goods inventory budget
 - i) A selling and administrative budget
2. A cash budget, by quarter and in total
3. A budgeted income statement for the four- quarter ending December 31, 20x4
4. A budgeted balance sheet as of December 31, 20x4.

Solution-Preparation of Master Budget (Manufacturing Company)

1. a) Sales Budget

Quarter				
	1	2	3	4
Expected sales in units	10, 000	30, 000	40, 000	20, 000
Selling price per unit	x Br. 20	x Br. 20	x Br.20	x Br.20
Total sales	<u>Br.200, 000</u>	<u>Br.600, 000</u>	<u>Br.800, 000</u>	<u>Br.400, 000</u>

b) Schedule of Expected Cash Collections

Quarter					
	1	2	3	4	Total

30% of the previous quarter sales	Br. 90, 000	Br.60, 000	Br.180, 000	Br.240, 000	Br.570, 000
70% of the current quarter sales	<u>140, 000</u>	<u>420, 000</u>	<u>560, 000</u>	<u>280, 000</u>	<u>1, 400, 000</u>
Total collections	<u>Br.230, 000</u>	<u>Br.480, 000</u>	<u>Br.740, 000</u>	<u>Br. 520, 000</u>	<u>Br.1, 970, 000</u>

c) Production Budget

After the sales budget has been prepared, the production requirements for the forth-coming budget period can be determined and organized in the form of a production budget. Sufficient goods will have to be available to meet sales and provide for the desired ending inventory. A portion of these goods will already exist in the form of a beginning inventory. The remainder will have to be produced. Therefore, production needs can be determined as follows:

Budgeted sales in units XXXX
Add desired ending inventory XXXX
Total needs..... XXXX
Less beginning inventory..... XXXX
Required production..... .XXXX

The schedule given below shows the production budget for Great Company. Note that the desired level of the ending inventory influences production requirements for a quarter. Inventories should be carefully planned. Excessive inventories tie up funds and create storage problems. Insufficient inventories can lead to lost sales or crash production efforts in the following period

	Quarter				Total
Expected sales(units)	10, 000	30, 000	40, 000	20, 000	100, 000
Add: Desired Ending Inventory	<u>6, 000</u>	<u>8, 000</u>	<u>4, 000</u>	<u>3, 000</u>	<u>3, 000</u>
Total needs	16, 000	38, 000	44, 000	23, 000	103, 000
Lees: Beginning Inventory	<u>2, 000</u>	<u>6, 000</u>	<u>8, 000</u>	<u>4, 000</u>	<u>2, 000</u>
Units to be produced	<u>14, 000</u>	<u>32,000</u>	<u>36, 000</u>	<u>19, 000</u>	<u>101, 000</u>
	1	2	3	4	

d) Direct Materials Budget

Returning to Great Company's budget data, after the production requirements have been computed, a direct materials budget can be prepared. The direct materials budget details the raw materials that must be purchased to

fulfill the production budget and to provide for adequate inventories. The required purchases of raw materials are computed as follows:

Raw materials needed to meet the production schedule.....XXXX
 Add desired ending inventory of raw materials.....XXXX
 Total raw materials needs..... .XXXX
 Less beginning inventory of raw materials..... XXXX
Raw materials to be purchased..... .XXXX

	Quarter				Total
	1	2	3	4	
Production needs(pounds)	210, 000	480, 000	540, 000	285, 000	1, 515, 000
Add: Desired ending inventory	<u>48, 000</u>	<u>54, 000</u>	<u>28, 500</u>	<u>22, 500</u>	<u>22, 500</u>
Total needs	258, 000	534, 000	568, 500	307, 500	1, 537, 500
Less: Beginning inventory	<u>21, 000</u>	<u>48, 000</u>	<u>54, 000</u>	<u>28, 500</u>	<u>21, 000</u>
Raw materials to be purchased(pounds)	<u>237, 000</u>	<u>486, 000</u>	<u>514, 500</u>	<u>279, 000</u>	<u>1, 516,500</u>

Raw Materials to be purchased (in birrs)

	1	2	3	4	Total
Raw materials to be purchased	237, 000	486, 000	514, 500	279, 000	1, 516, 500
Raw materials cost per pound	<u>x Br.0.20</u>	<u>x Br.0.20</u>	<u>x Br.0.20</u>	<u>x Br.0.20</u>	<u>x Br.0.20</u>
Total	<u>Br.47, 400</u>	<u>Br.97, 200</u>	<u>Br.102, 900</u>	<u>Br.55, 800</u>	<u>Br.303, 300</u>

e) Schedule of Expected Cash Disbursements (for Materials Purchase)

	Quarter				Total
	1	2	3	4	
50% of the previous quarter	Br. 25, 800	Br.23, 700	Br.48, 600	Br.51, 450	Br.149, 550
50% of the current quarter	<u>23, 700</u>	<u>48, 600</u>	<u>51, 450</u>	<u>27, 900</u>	<u>151, 650</u>
Total cash disbursement	<u>Br.49, 500</u>	<u>Br.72, 300</u>	<u>Br.101, 050</u>	<u>Br.79, 350</u>	<u>Br.301, 200</u>

f) Direct Labor Budget

The direct labor budget is also developed from the production budget. Direct labor requirements must be computed so that the company will know whether sufficient labor time is available to meet production needs. By

knowing in advance just what will be needed in the way of labor time throughout the budget year, the company can develop plans to adjust the labor force as the situation may require. Firms that neglect to budget run the risk of facing labor shortage or having to hire and lay off at awkward times. Erratic labor policies lead to insecurity and inefficiencies on the part of employees. To compute direct labor requirements, the number of units of finished product to be produced each period (month, quarter, and so on) is multiplied by the number of direct labor-hours required to produce a single unit. Many different types of labor may be involved. If so, then the computation should be by type of labor needed. The labor requirements can then be translated into expected direct labor costs. How this is done will depend on the labor policy of the firm. In schedule given below, the management of Great Company has assumed that the direct labor force will be adjusted as the work requirement change from quarter to quarter (for example as units produced changes from 14, 000 units in quarter 1 to 32, 000 units in quarter 2 for Great Company, the direct labor work force will be fully adjusted to the workload, i.e., total hours of direct labor time needed) . In that case, the total direct labor cost is computed by simply multiplying the direct labor-hour required by the direct labor rate hour as was done in the schedule here under.

	Quarter				Total
	1	2	3	4	
Direct labor time needed	11, 200	25, 600	28, 800	15, 200	80, 800
Direct labor cost per hour	<u>x Br.7.50</u>	<u>x Br.7.50</u>	<u>x Br.7.50</u>	<u>x Br.7.50</u>	<u>x Br.7.50</u>
Total direct labor cost	<u>Br.84, 000</u>	<u>Br.192, 000</u>	<u>Br.216, 000</u>	<u>Br.114, 000</u>	<u>Br.606, 000</u>

g) Manufacturing Overhead (MOH) Budget

The manufacturing overhead budget provides a schedule of all costs of production other than direct materials and direct labor. These costs should be broken down by cost behavior for budgeting purposes and a predetermined overhead rate developed. This rate will be used to apply manufacturing overhead to units of product throughout the budget period.

A computation showing budgeted cash disbursement for manufacturing overhead should be made for use in developing the cash budget. Since some of the overhead costs do not represent cash outflows, the total budgeted manufacturing overhead costs must be adjusted to determine the cash disbursement for manufacturing overhead. At Great Company, the only significant noncash manufacturing overhead cost is

depreciation. Any depreciation charges included in manufacturing overhead must be deducted from the total in computing expected cash payments, since depreciation is a noncash charge.

	Quarter				Total
	1	2	3	4	
Variable overhead	Br.22, 400	Br.51, 200	Br.57, 600	Br.30, 400	Br.161,600
Fixed overhead	<u>60, 600</u>	<u>60, 600</u>	<u>60, 600</u>	<u>60, 600</u>	<u>242,400</u>
Total MOH	Br.83, 000	Br.111, 800	Br.118, 200	Br.91, 000	Br.404,000
Less: Depreciation	<u>15, 000</u>	<u>15, 000</u>	<u>15, 000</u>	<u>15, 000</u>	<u>60, 000</u>
Cash disbursements for MOH	<u>Br.68, 000</u>	<u>Br.96, 800</u>	<u>Br.103, 200</u>	<u>Br.76, 000</u>	<u>Br.344, 000</u>

h) Ending Finished Goods Inventory Budget

After completing schedules (a) to (g), the company had all of the data needed to compute unit product costs. This computation was needed for two reasons: first, to know how much to charge as cost of goods sold on the budgeted income statement; and second, to know what amount to put on the balance sheet inventory account for unsold units. The carrying cost of the unsold units is computed on the ending finished goods inventory budget

Budgeted Finished Goods Inventory 3, 000

Unit product cost Br.13

Ending Finished Goods Inventory in birrs Br.39, 000

Production cost per unit

	<u>Quantity (unit)</u>	<u>Cost</u>	<u>Total</u>
Direct materials	15 pounds	Br.0.20 per pound	Br.3
Direct labor	0.8 hours	7.50 per hour	6
Manufacturing overhead	0.8 hours	5.00 per hour	<u>4</u>
Unit product cost			<u>Br.13</u>

MOH rate= $\frac{\text{Total MOH}}{\text{Direct labor hours}} = \frac{404, 000}{80, 800} = \text{Br.5.00}$

Direct labor hours 80, 800

i) Selling and Administrative Expenses Budget

	Quarter	Total
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	1	2	3	4	
Variable selling expenses	Br.18, 000	Br.54, 000	Br.72, 000	Br.36, 000	Br.180, 000
Fixed selling & administrative expenses					
Advertising	20, 000	20, 000	20, 000	20, 000	80, 000
Executive salaries	55, 000	55, 000	55, 000	55, 000	220, 000
Insurance	-	1, 900	37, 750	-	39, 650
Property taxes	-	-	-	18,150	18,150
Depreciation	<u>10, 000</u>	<u>10, 000</u>	<u>10, 000</u>	<u>10, 000</u>	<u>40, 000</u>
Total budgeted selling & administrative expenses	<u>Br.103, 000</u>	<u>Br.140, 900</u>	<u>Br.194, 750</u>	<u>Br.139, 150</u>	<u>Br.577, 800</u>

Disbursement for Selling & Administrative Expenses

	Quarter				Total
	1	2	3	4	
Budgeted Selling & Administrative	Br.103, 000	Br.140, 900	Br.194, 750	Br.139, 150	Br.577, 800
Less: Depreciation	<u>10, 000</u>	<u>10, 000</u>	<u>10, 000</u>	<u>10, 000</u>	<u>40, 000</u>
Total Cash Disbursements	<u>Br.93, 000</u>	<u>Br.130, 900</u>	<u>Br.184, 750</u>	<u>Br.129, 150</u>	<u>Br.537, 800</u>

2. a) Cash Budget

	Quarter				Total
	1	2	3	4	
Cash balance, beginning	Br.42, 500	Br.40, 000	Br.40, 000	Br.40, 500	Br.42, 500
Add : Collection from customers	<u>230, 000</u>	<u>480, 000</u>	<u>740, 000</u>	<u>520, 000</u>	<u>1, 970, 000</u>
Total cash available before financing	<u>272, 500</u>	<u>520, 000</u>	<u>780, 000</u>	<u>560, 500</u>	<u>2, 012, 500</u>
Less: Disbursements for					
Direct materials	49, 500	72, 300	100,050	79, 350	301,200
Direct labor	84, 000	192, 000	216,000	114, 000	606,000
Manufacturing overhead	68, 000	96, 800	103,200	76, 000	344,000
Selling & Administrative	93, 000	130, 900	184,750	129, 150	537,800
Equipment purchases	50, 000	40, 000	20,000	20,000	130,000
Dividend	<u>8, 000</u>	<u>8, 000</u>	<u>8, 000</u>	<u>8, 000</u>	<u>32,000</u>
Total disbursements	352, 500	540,000	632,000	426,500	1,951,000
Minimum cash balance	<u>40, 000</u>	<u>40, 000</u>	<u>40, 000</u>	<u>40, 000</u>	<u>40, 000</u>
Total need	<u>392, 500</u>	<u>580, 000</u>	<u>672, 000</u>	<u>466, 500</u>	<u>1, 991,000</u>
Excess (deficiency) of cash available over total need	<u>(120, 000)</u>	<u>(60, 000)</u>	<u>108, 000</u>	<u>94, 000</u>	<u>21, 500</u>
Financing:					
Borrowing(at beginning)	120,000	60, 000	-	-	180, 000
Repayments(at ending)	-	-	(100, 000)	(80,000)	(180,000)
Interest(at 10% per annum)	-	-	<u>(7,500)</u>	<u>(6,500)</u>	<u>(14,000)</u>
Total financing	<u>120, 000</u>	<u>60, 000</u>	<u>(107,500)</u>	<u>(86,500)</u>	<u>(14,000)</u>
Cash balance, ending	<u>Br.40,000</u>	<u>Br.40, 000</u>	<u>Br.40, 500</u>	<u>Br.47, 500</u>	<u>Br.47, 500</u>

b) Budgeted Income Statement

Great Company

Budgeted Income Statement
For the Year Ended December31, 20x4

Sales [100, 000units at Br.20 Schedule 1(a)]	Br.2, 000, 000
Cost of Goods Sold [100, 000 units at Br.13 Schedule1 (h)]	<u>1, 300, 000</u>
Gross Margin	700, 000
Selling & Administrative Expenses [Schedule 1 (i)]	<u>577, 800</u>
Net Operating Income	122, 200
Interest Expense [Schedule 2(a)]	<u>14, 000</u>
Net Income	<u>Br. 108, 200</u>

c) Budgeted Balance Sheet

Great Company
Budgeted Balance Sheet
December31, 20x4

ASSETS

Current assets:

Cash [Schedule 2(a)]	Br. 47, 500
Accounts Receivable	120, 000
Raw Materials Inventory	4, 500
Finished Goods Inventory	<u>39, 000</u>

Total current assets Br.211, 000

Plant and Equipment:

Land	Br.80, 000
Building and Equipment	830, 000
Accumulated Depreciation	<u>(392, 000)</u>

Plant and Equipment, net 518, 000

Total assets **Br.729, 000**

Liabilities and Stockholders' Equity

Current liabilities:

Accounts payable (raw materials) Br.27, 900

Stockholders' equity:

Common stock, no par	Br.175, 000
Retained earnings	<u>526, 100</u>

Total stockholders' equity 701, 100

Total liabilities and stockholders' equity **Br.729, 000**

Preparation of Master Budget (Merchandising Company)

Example 1: Blue Nile Company's newly hired accountant has persuaded management to prepare a master budget to aid financial and operating decisions. The planning horizon is only three months, January to March. Sales in December (20x3) were Br. 40, 000. Monthly sales for the first four months of the next year (20x4) are forecasted as follows:

January	Br. 50, 000
February	80, 000
March	60, 000
April	50, 000

Normally 60% of sales are on cash and the remainders are credit sales. All credit sales are collected in the month following the sales. Uncollectible accounts are negligible and are to be ignored.

Because deliveries from suppliers and customer demand are uncertain, at the end of any month Blue Nile wants to have a basic inventory of Br. 20, 000 plus 80% of the expected cost of goods to be sold in the following month. The cost of merchandise sold averages 70% of sales. The purchase terms available to the company are net 30 days. Each month's purchase are paid as follows:

- 50% during the month of purchase and,
- 50% during the month following the purchases

Monthly expenses are:

- Wages and commissions.....Br. 2, 500 + 15% of sales, paid as incurred.
- Rent expense.....Br. 2, 000 paid as incurred.
- Insurance expense.....Br.200 expiration per month
- Depreciation including truck.....Br.500 per month
- Miscellaneous expense.....5% of sales, paid as incurred.

In January, a used truck will be purchased for Br. 3, 000 cash. The company wants a minimum cash balance of Br. 10, 000 at the end of each month. Blue Nile can borrow cash or repay loans in multiples of Br. 1, 000. Management plans to borrow cash more than necessary and to repay as promptly as possible. Assume that the borrowing takes place at the beginning, and repayment at the end of the months in question. Interest is paid when the related loan is repaid. The interest rate is 18% per annum. The closing balance sheet for the fiscal year just ended at December 31, 20x3, is:

Blue Nile Company

Balance Sheet
December 31, 20x3

ASSETS

Current assets:

Cash	Br. 10, 000	
Account receivable	16,000	
Merchandise inventory	48, 000	
Unexpired insurance	<u>1, 800</u>	Br.75, 800

Plant assets:

Equipment, fixture and other	Br.37, 000	
Accumulated depreciation	<u>12, 800</u>	Br.24, 200

Total assets

Br.100,000

LIABILITIES AND OWNERS' EQUITY

Liabilities:

Accounts payable	Br.16, 800	
Accrued wages and commissions payable	<u>4, 250</u>	Br.21, 050

Capital:

Owners' equity		<u>78, 950</u>
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Total liabilities and owners' equity

Br.100, 000

Instructions:

1) Using the data given above, prepare the following detailed schedules for the first quarter of the year:

- b) Sales budget
- c) Cash collection budget
- d) Purchase budget
- e) Disbursement for purchases
- f) Operating expenses budget
- g) Disbursement for operating expenses

2) Using the budget data given above and the schedules you have prepared, construct the following pro forma financial statements

- a. Income statement for the first quarter of the year.
- b. Cash budget including receipts, payments, and effect of financing
- c. Balance sheet at March 31, 20x3.